

The Impact of AI Usage on Student Burnout Risk

A Supervised Classification Analysis

This report applies supervised machine learning classification techniques to predict student burnout risk based on patterns of generative-AI usage and study behavior. It documents the dataset, modeling approach, model comparison and selection, key findings, and limitations, with the goal of supporting data-driven decisions around AI usage policy in academic settings.

1. Dataset Selection and Description

The dataset used in this analysis, `ai_student_impact_dataset.csv`, is a publicly available dataset sourced from Kaggle, focused on the relationship between generative-AI usage and academic/wellbeing outcomes among students. It contains **50,000 student records** across **16 features**, with no missing values in any column.

Target variable

The classification target is **Burnout_Risk_Level**, a three-class categorical outcome (Low / Medium / High) representing a student's assessed risk of academic burnout. The classes are moderately imbalanced:

Class	Count	Share of dataset
Medium	21,144	42.3%
Low	16,369	32.7%
High	12,487	25.0%

Key features

After excluding the non-predictive identifier column (`Student_ID`), the dataset offers a mix of behavioral, academic, and attitudinal features relevant to the classification problem:

- AI usage behavior:** `Weekly_GenAI_Hours`, `Tool_Diversity`, `Primary_Use_Case`, `Prompt_Engineering_Skill`, `Paid_Subscription`
- Attitudes toward AI:** `Perceived_AI_Dependency`, `Institutional_Policy`
- Academic performance and effort:** `Pre_Semester_GPA`, `Post_Semester_GPA`, `Traditional_Study_Hours`, `Skill_Retention_Score`
- Student context:** `Major_Category`, `Year_of_Study`
- Wellbeing indicator:** `Anxiety_Level_During_Exams`

This feature set is directly relevant to the classification problem: it combines measures of how intensively and dependently students use generative AI tools with independent measures of academic performance and exam-related anxiety, allowing the model to test whether AI usage patterns are predictive of burnout risk rather than assuming the relationship a priori.

2. Analytical Approach and Model Development

Exploratory data analysis

The dataset was confirmed to contain no missing values. Feature types were categorized into continuous numerical features (e.g. GPA, study hours), discrete/ordinal numerical features recorded as integers (Tool_Diversity, Perceived_AI_Dependency, Anxiety_Level_During_Exams), and categorical (string) features. Distribution plots showed that all continuous numerical features were right- or left-skewed rather than normally distributed, and ranges varied substantially between features (e.g. GPA on a 0-4 scale vs. weekly AI hours on a 0-40 scale), confirming the need for both a skewness correction and standardization prior to modeling.

Data preparation

- **Splitting:** a 75/25 train/test split was used, stratified on the target variable to preserve class proportions given the observed imbalance (random_state fixed for reproducibility).
- **Scaling:** continuous numerical features were first passed through a Yeo-Johnson power transform to reduce skew, then standardized (zero mean, unit variance) with StandardScaler, fit on the training set only and applied to the test set.
- **Encoding:** nominal categorical features (Major_Category, Primary_Use_Case, Paid_Subscription, Institutional_Policy) were one-hot encoded to avoid imposing a false ordinal relationship; features with a genuine natural order (Year_of_Study, Prompt_Engineering_Skill, and the target itself) were ordinal-encoded to preserve that order.
- **Feature engineering / selection:** a correlation analysis against the encoded target was used post-modeling to assess feature importance and test a reduced feature set (see Section 4).

Model comparison

Six classification algorithms were trained on identical preprocessed data and compared using weighted F1 score (weighted average was used, rather than macro, to reflect the natural class distribution while still accounting for imbalance across the three classes) on the held-out test set:

Model	Weighted F1 (default hyperparameters)
Logistic Regression	0.53 — best
Support Vector Classifier (SVC)	< 0.53
Decision Tree	< 0.53
K-Nearest Neighbors	< 0.53
Random Forest	< 0.53

Extra Trees	< 0.53
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Logistic Regression achieved the highest weighted F1 score (0.53) among the six default-parameter models and was carried forward for hyperparameter tuning.

Hyperparameter tuning

A grid search with 5-fold cross-validation was run over Logistic Regression's regularization type (L1 / L2 / Elastic Net), inverse regularization strength C, elastic-net mixing ratio, and solver, optimizing for weighted F1. Invalid solver/penalty combinations were scored as zero rather than raising an error, allowing the search to run over the full grid safely. The best configuration found was:

Parameter	Best value
Penalty	L2
C	0.5
Solver	lbfgs
l1_ratio	0.3 (unused — only applies to elasticnet)

Notably, the tuned model did not outperform the default-parameter Logistic Regression baseline (weighted F1 remained 0.53), indicating that hyperparameter choice was not the limiting factor for this model (see Section 5).

3. Model Selection and Justification

Logistic Regression was selected as the final model. This choice was based on the following considerations:

- **Predictive performance:** it matched or exceeded every other model tested, including more flexible non-linear alternatives (Random Forest, Extra Trees, KNN, SVC, Decision Tree), under identical preprocessing and evaluation conditions.
- **Interpretability:** as a linear model, its coefficients can be directly related to feature influence on each class, which is valuable for communicating findings to non-technical stakeholders such as academic staff or policy decision-makers.
- **Robustness:** Logistic Regression is less prone to overfitting on a dataset with a large number of one-hot encoded, low-signal features than higher-variance models such as Decision Trees or KNN, which showed no advantage here despite their added complexity.
- **Alignment with stakeholder needs:** given the objective is to support institutional decision-making (e.g. AI usage guidance), a transparent, easily explainable model is preferable to a marginally different but opaque alternative — and in this case there was no performance trade-off to accept, since Logistic Regression was already the top performer.

It is worth noting explicitly that the performance gap between all six candidate models was narrow. This is treated as an important finding in itself in Section 5: it suggests the ceiling on predictive performance is set primarily by the information content of the dataset, not by model choice.

4. Key Findings and Insights

Final model performance

On the held-out test set (12,500 students), the tuned Logistic Regression model achieved an overall accuracy of 53% and a weighted F1 score of 0.53:

Burnout risk class	Precision	Recall	F1-score	Support
Low	0.55	0.44	0.49	4,092
Medium	0.48	0.61	0.54	5,286
High	0.63	0.51	0.56	3,122
Weighted avg.	0.54	0.53	0.53	12,500

As a reference point, a naive baseline that always predicts the majority class (Medium) achieves 42% accuracy on this data. The model's 53% accuracy represents a real, if modest, improvement over that baseline — the model is extracting genuine signal from the features, but that signal is limited. The model identifies the High-risk class most precisely (0.63 precision) but with the lowest recall, meaning it correctly flags High-risk students less than it misses them — a relevant caveat given that missed High-risk cases carry the highest practical cost in a real intervention setting.

Feature importance

A correlation analysis between each (encoded) feature and the target variable was used to assess feature importance. Two features stand out substantially above the rest:

Feature	Correlation with burnout risk
Weekly_GenAI_Hours	+0.44
Perceived_AI_Dependency	+0.37
Anxiety_Level_During_Exams	+0.16
Year_of_Study	+0.16
Traditional_Study_Hours	-0.13
All remaining 20 features	each below 0.10

A follow-up experiment retrained the final model on only the eight features with the strongest correlations. This did not change performance (accuracy and weighted F1 remained 0.53), confirming that the remaining, weakly-correlated features were not meaningfully contributing to — or detracting from — the model's predictions.

Practical implications

The two dominant predictors — weekly hours of generative-AI use and self-reported AI dependency — both show a positive relationship with burnout risk. This supports the project's objective of informing data-driven guidance: institutions and instructors have a measurable, actionable basis for cautioning students about heavy or dependent AI usage patterns, both for academic outcomes and for wellbeing, rather than relying on anecdotal concern alone.

5. Limitations and Future Work

Limitations

- **Weak signal in most features:** 20 of 25 encoded features showed a correlation below 0.10 with the target. Given tuning did not move performance and every model architecture converged to a similar ceiling, the limitation most likely lies in the information content of the dataset rather than the modeling approach.
- **Class imbalance:** although addressed via stratified sampling during the train/test split, the models were not trained with class-weighting or resampling (e.g. SMOTE), which may partly explain the comparatively lower recall on the minority High-risk class — the class where recall arguably matters most.
- **Linear model assumption:** the selected model assumes linear (additive) relationships between features and the log-odds of each class. Non-linear models (Random Forest, Extra Trees, SVC, KNN) were tested only with default hyperparameters and were not individually tuned before being ruled out — it is possible a tuned non-linear model could capture interaction effects (e.g. between AI usage and traditional study hours) that a linear model cannot.
- **Single train/test split:** model comparison in the selection stage relied on one stratified split rather than cross-validated scores, so small differences between the candidate models' F1 scores should be interpreted cautiously.
- **Correlation-based feature importance:** the feature importance analysis relied on linear (Pearson-style) correlation with the encoded target, which does not capture non-linear or interaction effects between features and burnout risk.
- **Dataset provenance:** as a public Kaggle dataset, the underlying data-generating process (and whether it reflects a real institutional survey or a synthetically generated dataset) is not fully verifiable, which affects how strongly the findings should be generalized to real-world populations.

Future work

- Apply class-weighting or resampling techniques (e.g. SMOTE, `class_weight='balanced'`) specifically to improve recall on the High-risk class, given its practical importance.
- Tune the non-linear candidate models (Random Forest, Extra Trees, SVC) rather than comparing them only at default settings, to confirm whether the current performance ceiling is truly architecture-independent.
- Engineer interaction features — for example, a ratio or difference between `Weekly_GenAI_Hours` and `Traditional_Study_Hours` — to test whether burnout risk depends on the balance between AI-assisted and traditional study rather than either measure alone.
- Validate model comparison and selection using cross-validated scores rather than a single held-out split, for more reliable model ranking.
- If possible, validate findings against an independent or institutionally-sourced dataset to confirm the relationship between AI dependency and burnout risk generalizes beyond this particular sample.